TEST DATE : _____ Take Home Practice Test will be given out soon!

Name:_____ Period:____

Unit 2 Test Review: Systems of Matter and Energy

Use this review to get ready for the test:
☐ Read each topic. Underline anything you don't remember and look it up in your notes.
☐ Open your Reference Tables to the pages listed and find each chart.
☐ Practice the skills on page 2 with your practice test.

Part 1: What you need to know

TOPIC	Key ideas	Reference Tables
Heat transfer	**Radiation** = waves through space (the Sun). **Conduction** = things touching. **Convection** = a moving liquid or gas; warm rises, cool sinks.	
Energy sources	The **Sun** drives surface processes (weathering, erosion, water cycle). **Earth's hot interior** drives melting, metamorphism and mantle convection.	
Density	Denser material sinks; less dense material rises. Ice is less dense than water, so water expands when it freezes.	
Phase changes	Melting and evaporation **absorb** energy. Freezing and condensation **release** energy. Temperature stays the same during a phase change.	
Minerals	Identify by properties: luster, hardness (glass = 5.5), streak, cleavage or fracture, acid reaction. Location, size and mass do not identify a mineral.	Mineral Identification Flowchart (pp. 16–17)
Igneous rocks	Slow cooling underground = large crystals. Fast cooling at the surface = small crystals. **Felsic** = light (quartz, K-feldspar). **Mafic** = dark (pyroxene, olivine).	Igneous rocks and Bowen's Reaction Series (p. 14)
Sedimentary rocks	Weathering → erosion → deposition → compaction and cementation. Chemical rocks form when dissolved minerals precipitate (limestone, rock salt).	Rock Cycle Infographic (p. 15)
Metamorphic rocks	Heat and pressure change a rock **without melting** it. Banded or layered = foliated (slate, gneiss). No layers = nonfoliated (marble, quartzite).	Rock Cycle Infographic (p. 15)
Carbon cycle	CO_2 + rain → weak acid → weathers rock → ions to ocean → limestone → melted → volcanoes release CO_2.	

Command words
Identify = name it. **Describe** = tell what happens. **Explain** = tell why or how.
Refute = prove it wrong with evidence. **Relative** = compare (more or less).

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Part 2: Regents skills and how to approach them

SKILL	How to approach it	Hint
Read the set	Read the title and every visual first: title, key, labels, axes, units.	Many answers are right in the reading.
“Which row” tables	Check one column at a time. Cross out rows as you go.	One wrong box = the whole row is wrong.
Sequence steps	Find the start and the end. Put the steps in order so each one causes the next.	Reread your order like a story.
Read a graph	As _____ increases, _____ (increases / decreases / stays the same).	Check which way each axis goes. A flat line = no change.
Use a model	The material starts at _____, moves by _____ and becomes _____.	Trace the arrows first.
Draw on a model	Draw exactly what is asked. For convection, hot material rises and cool material sinks.	Each arrow points one clear way.
Pick the evidence	Cross out clues that can’t answer the question, like where it was found or how big it is.	Use only clues that match the chart.
Use the Reference Tables	Match the question to the page: p. 14, p. 15 or pp. 16–17.	Some questions need two pages.
Write a 1-credit answer	Name the mineral, process or number. Answer every part.	“Felsic” or “it changes” alone is not enough.

Sentence starters

WORD	Start with
Identify	The _____ is _____.
Describe	First _____, then _____.
Explain	_____ happens because _____.
Refute	The claim is wrong because _____.
Relative	_____ has more/less _____ than _____.

STUCK? Use this checklist:
☐ Reread the question and circle the command word
☐ Underline key words and numbers
☐ Look again at the visuals and find the right Reference Tables page
☐ Check that you answered every part